



Roll No. 5.019932.....

97764

**BCA 4 Years 3rd Semester (NEP)
Examination – December, 2025**

DATABASE MANAGEMENT SYSTEM

Paper : 24BCA403DS03

Time : Three hours]

[Maximum Marks : 50

34

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt *five* questions in all, selecting *one* question from each Unit. Question No. 1 is *compulsory*. All questions carry equal marks.

1. (a) Differentiate between primary key, candidate key and super key.

5 × 2 = 10

(b) Define data, database and DBMS.

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P. T. O.



- (c) What do you mean by database backup ?
- (d) Discuss the state diagram of transaction.
- (e) Why are duplicate tuples not allowed in a relation ?

UNIT – I

2. (a) Construct an E-R diagram for a university system. 7
- (b) Discuss different views of data. 3
3. (a) List the significant differences between DBMS and file system. 3
- (b) What do you mean by database model ? Discuss various types of database models. 7

UNIT – II

4. Explain the following operations in relational algebra by giving suitable examples : 10
- (a) SELECT
- (b) PROJECT
- (c) PRODUCT



(d) JOIN

(e) UNION

5. Write note on :

(a) Views

5

3 1/2

(b) Relational model constraints

5

3 1/2

UNIT – III

6. What do you mean by normalization ? Explain in 1NF, 2NF, 3NF, and BCNF with examples.

10



7. What is the two-phase locking protocol ? How does it guarantee serializability ?

10

UNIT – IV

8. What is a file ? Discuss sequential file organization with suitable examples.

10

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(3)

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9. Write notes on :

(a) Failure classification

5

3

(b) Remote Backup System

5

3

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